

## Summarization and Simplification of Medical Articles using Natural Language Processing

**Problem Statement:** The paper addresses the challenge of improving *health literacy* by easing access to complex healthcare information by summarising medical texts and simplifying them lexically by translating specific medical terminology to laymen's terms.

**Algorithms Used:** Used 4 pretrained transformer-based models - *ALBERT*, *DistilBERT*, *SciBERT*, *GPT-2* for extracting summaries from articles and another pretrained model - *en\_ner\_bionlp13cg\_md* for Named Entity Recognition of the medical terms. Used *Lemmatization* for each medical entity and finally *Web Scraping* to generate a simplified meaning.

**Datasets:** Dataset of 400 medical articles taken from the *Cochrane Database of Systematic Reviews (CDSR)*.

**Model Training and Testing:** In generated summary, the medical entities present in the summary are highlighted in green and their respective meaning is displayed to the user when he hovers over the highlighted word. *ROUGE* i.e. *Recall-Oriented Understudy for Gisting Evaluation* metric has been used to evaluate the summaries produced by the models.

**Results:** The compression rate of the models is 40%. Below figure shows a comparative analysis of the ROUGE scores obtained by the four chosen pretrained models.

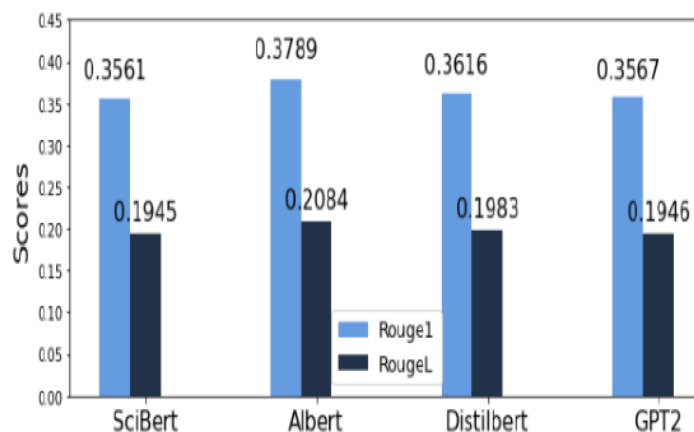


Fig. 5. Comparative analysis of the ROUGE-1 and ROUGE-L scores

**Conclusions:** Authors employ an extractive text summarization approach using ALBERT. Lexical simplification of the summary is done using Named Entity Recognition along with simplifying the complex words through web scraping.

**Open Questions:** The paper suggests future work on improving the paragraph summaries and identification of non-medical complex words and medical abbreviations.

**Relevance to Our Team:** The evaluations into the various transformer based models can guide our team in selecting appropriate models for text summarization which can help clinical researchers to quickly navigate and review documents.

## **Extracting Health Evidence Information from Biomedical Literature using LLMs**

**Problem Statement:** This article addresses the challenge of bridging the gap between research findings and real-world healthcare applications by exploring *LLMs* to enhance medical practice efficiency and advance *evidence based medicine*.

### **Algorithms Used:**

- *INSTRUCTOR* and *OpenAI* for embeddings
- *FAISS* for storing vector embeddings in a knowledge base
- *PICO* Extraction Algorithm(User-Defined)
- *Stuff* for document processing
- *Streamlit* Framework to design an interface for user engagement
- *LangChain* for summarising complex PDFs
- *RAG*-based pipeline
- *RAGAs* for evaluation of retriever and LLMs
- OpenAI LLMs: *GPT-3.5 turbo*, *GPT-4*, *GPT-4 turbo*, and *GPT-4o* to generate responses

### **Datasets:**

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3952010/>  
<https://link.springer.com/article/10.1208/s12249-021-02058-y>  
[https://www.thelancet.com/journals/landia/article/PIIS2213-8587\(22\)00387-4/abstract](https://www.thelancet.com/journals/landia/article/PIIS2213-8587(22)00387-4/abstract)  
<https://onlinelibrary.wiley.com/doi/full/10.1002/pst.2020>  
<https://www.sciencedirect.com/science/article/pii/S0753332223014488>

**Model Training and Testing:** Overall, 12 PICO queries were generated from the documents. These queries were then passed to each LLM to extract evidence using the PICO framework and answer the queries with the extracted evidence.

**Results:** The average performance of each LLM is detailed in below figure:

| Model         | Embeddings | AVG Context Recall | AVG Context Precision | AVG Faithfulness | AVG Answer relevancy |
|---------------|------------|--------------------|-----------------------|------------------|----------------------|
| GPT 3.5 turbo | Instructor | 75.0%              | 95.5%                 | 93.7%            | 82.4%                |
| GPT 3.5 turbo | OpenAI     | 75.0%              | 99.7%                 | 85.00%           | 82.3%                |
| GPT 4         | Instructor | 85.0%              | 95.0%                 | 96.2%            | 85.0%                |
| GPT 4         | OpenAI     | 75.0%              | 100%                  | 96.7%            | 88.3%                |
| GPT 4 turbo   | Instructor | 71.0%              | 94.0%                 | 95.2%            | 87.6%                |
| GPT 4 turbo   | OpenAI     | 90.0%              | 99.7%                 | 89.6%            | 87.8%                |
| GPT 4o        | Instructor | 71.0%              | 94.2%                 | 76.3%            | 85.2%                |
| GPT 4o        | OpenAI     | 75.0%              | 99.7%                 | 72.67%           | 87.2%                |

### **Conclusions:**

- LLMs like GPT-4 and GPT-4 Turbo can effectively handle PICO-based queries in systematic medical reviews.
- The choice of embeddings significantly influences performance.

**Open Questions:** The predefined limits on context retrieval can result in incomplete information. Future work should aim to increase the context size to take full advantage of models like GPT-4 Turbo, even though this may come with higher costs which can be addressed by their fine-tuning.

**Relevance to Our Team:** This article is very relevant for our project as it caters to all the aspects of our objective and presents some challenges which we can work on.

## Completing A Systematic Review in Hours instead of Months with Interactive AI Agents

**Problem Statement:** This article addresses accuracy, relevance and quality issues with existing automatic summarization methods due to lack of domain expertise and intervention. It introduces a human-centered interactive AI agent *InsightAgent* powered by LLMs for SR workflow. This agent also provides visualizations of the corpus and agent trajectories, allowing users to monitor the actions of the agent and provide real-time feedback.

### Algorithms Used:

- *RSS map* for visualizing corpus structure
- *K-means clustering* for corpus partitioning into semantically distinct clusters
- LLMs: *GPT-4o*, *Llama 3.3 70B*
- *VA* based interface

**Datasets:** Literature corpus for 15 SRs from PubMed with size ranging from 72 to 7,356 articles

**Model Training and Testing:** Insight agent is evaluated with both the LLMs in autonomous and user interaction modes. Also, comparison is done with existing agents like BM25, ChatCite and AutoSurvey. Evaluation is done by 9 medical experts on the basis of:

- **Record Screening** - Precision, Recall and F1 scores
- **SR Quality** – Rubric based method
- **User Experience** – Feedback Questionnaire and Interviews

**Results:** The average performance of each agent is detailed in below figures:

#### 4.2.1 Record Screening

#### Quality of the Generated Summaries

| System                                         | Recall (%) | Precision (%) | F1 % |
|------------------------------------------------|------------|---------------|------|
| BM25 (Top-100)                                 | 54.30      | 16.50         | 25.3 |
| ChatCite                                       | –          | –             | –    |
| AutoSurvey (Top-100)                           | 70.30      | 20.43         | 31.6 |
| <i>InsightAgent<sub>auto</sub></i> (Llama 3.3) | 66.40      | 62.40         | 64.3 |
| <i>InsightAgent</i> (Llama 3.3)                | 87.90      | 80.10         | 83.8 |
| <i>InsightAgent<sub>auto</sub></i> (GPT-4o)    | 71.10      | 51.90         | 60.0 |
| <i>InsightAgent</i> (GPT-4o)                   | 98.50      | 79.80         | 88.2 |

| System                                         | Score |
|------------------------------------------------|-------|
| ChatCite (GPT-4)                               | 47.1  |
| AutoSurvey (GPT-4o)                            | 54.0  |
| <i>InsightAgent<sub>auto</sub></i> (Llama 3.3) | 60.9  |
| <i>InsightAgent</i> (Llama 3.3)                | 70.2  |
| <i>InsightAgent<sub>auto</sub></i> (GPT-4o)    | 62.4  |
| <i>InsightAgent</i> (GPT-4o)                   | 79.7  |

| Question                            | <i>InsightAgent<sub>auto</sub></i> (Llama 3.3) | <i>InsightAgent</i> (Llama 3.3) | <i>InsightAgent<sub>auto</sub></i> (GPT-4o) | <i>InsightAgent</i> (GPT-4o) |
|-------------------------------------|------------------------------------------------|---------------------------------|---------------------------------------------|------------------------------|
| System was easy to use.             | 2.0/5                                          | 3.9/5                           | 2.1/5                                       | 4.0/5                        |
| Confidence in recommendations.      | 2.6/5                                          | 4.2/5                           | 2.8/5                                       | 4.5/5                        |
| Visualizations aided understanding. | 2.9/5                                          | 4.2/5                           | 2.9/5                                       | 4.2/5                        |
| Ability to guide or correct agents. | 2.7/5                                          | 3.9/5                           | 2.9/5                                       | 4.6/5                        |
| Overall satisfaction.               | 3.0/5                                          | 4.0/5                           | 3.2/5                                       | 4.3/5                        |

**Conclusions:** A single domain expert can finish a high quality SR in only 1.5 hours, rather than months, and reach 79.7% of human-written quality.

### Open Questions:

- It is a small scale study.
- Considering LLM cost and context length constraint, only title and abstract of each article is used in this study.
- Extraction of numerical statistics or weight evidence is missing.

**Relevance to Our Team:** This article focuses on rapid SRs reducing review time from days/weeks to hours which is one of the main objectives of our project.

## **Artificial intelligence to automate the systematic review of scientific literature**

**Problem Statement:** Manually preparing and writing a systematic literature review (SLR) takes considerable time and effort. This paper presents a survey of AI techniques proposed in the last 15 years to help researchers conduct systematic analyses of scientific literature.

### **Algorithms Used:**

- *Lingo3G* for document clustering (2013)
- *Fuzzy mining algorithm* to abstract different review models(2018)
- *Voting perceptron algorithm* for supervised learning based paper selection(2006)
- *BoW* representation of the papers(2006)
- *SVM light* for supervised learning based paper selection(2009)
- *TF-IDF (Term Frequency-Inverse Document Frequency)* metric to weight the importance of each word(2019)
- *Logistic Regression* for supervised learning based paper selection(2019)
- *Topic modelling algorithm* (Latent Dirichlet Allocation, LDA) for topic extraction(2019)
- *Ensemble Classifier* for supervised learning based paper selection(2021)
- *Platt scaling LR model* for supporting ensemble(2021)
- *Multinomial NB (MNB)* algorithm to use word count normalisation(2010)
- *Abstrackr* for active learning based paper selection(2012)
- *FASTREAD* for active learning based paper selection(2018)
- *FAST* for active learning based paper selection(2019)
- *DT (ID3 algorithm)* for inferring search strings for paper selection(2017)
- *BioBERT* for encoding sentences in reporting phase of SLR(2020)
- *Long short-term memory (LSTM)* recurrent neural network for transforming encoded sentences into summaries(2010)
- *LBJ* for automatic question generation for report evaluation(2007)

**Datasets:** 34 primary studies are selected from around 9027 references from ACM Library, IEEE Xplore, Scopus, SpringerLink and Web of Science.

**Model Training and Testing:** Not Applicable to this paper as this is a survey.

**Results:** NA

**Conclusions:** Findings reveal a clear interest in applying AI to support paper screening. Regarding other tasks like reporting, the number of studies in these areas is still low.

### **Open Questions:**

- One single task i.e. paper selection is predominant.
- AI techniques are still to be explored.
- Specialised algorithms can replace general purpose approaches.
- More active human involvement can benefit AI.
- End-users of SLR automation are not necessarily AI experts.
- AI-based automation of SLRs can be scaled up.
- Performance comparison between different methods and fields

**Relevance to Our Team:** This article focuses on early solutions and classical techniques used for automated SLR which would help us in survey of existing approaches.

## **Enhancing Context-Aware Search with Retrieval-Augmented Generation**

**Problem Statement:** To provide more accurate and contextually relevant search results from digital content, enhanced retrieval mechanisms are required. This paper explores how SLM-enhanced RAG architectures refine document search results and their implications for future developments.

### **Algorithms Used:**

- *BM25* for lexical matching
- *all-MiniLM-L6-v2* for sentence embeddings
- *FLAN-T5* for contextual ranking

**Datasets:** The corpus consists of single document, containing unstructured data.

*Document Title: Quantum Computing and its Impact on Cryptography*

*Content: Quantum computing harnesses quantum mechanics to perform computations at unprecedented speeds. One significant impact is on cryptographic security, where quantum algorithms like Shor's algorithm can break classical encryption schemes. Post-quantum cryptography is a developing field aiming to create quantum-resistant encryption methods.*

**Model Training and Testing:** Three retrieval models were evaluated: BM25, SLM, SLM+RAG on the query: "How does quantum computing affect encryption?"

**Results:** This experiment measures precision, recall, and response time to determine the most effective retrieval approach.

| Model                    | Relevance Score | Generated Response                                                                 |
|--------------------------|-----------------|------------------------------------------------------------------------------------|
| BM25 (Lexical Matching)  | 0.500           | N/A                                                                                |
| SLM (Semantic Matching)  | 0.7668          | N/A                                                                                |
| SLM + RAG (Hybrid Model) | 0.9202          | "Quantum algorithms like Shor's algorithm can break classical encryption schemes." |

### **Conclusions:**

- BM25 is effective for exact term matching and ranking documents based on term frequency and inverse document frequency but lacks semantic understanding.
- SLM Embeddings addresses many of BM25's limitations by employing dense vector representations that encode semantic meaning beyond keyword overlap and hence, improves retrieval accuracy. But it has high computational cost for large scale retrieval and reduced interpretability compared to BM25.
- SLM + RAG integrates retrieval and generative ranking to improve document relevance and contextual awareness, hence, mitigates hallucinations.

### **Open Questions:**

- Multi-Document Retrieval for Large-Scale Applications
- Hybrid approaches that integrate BM25 with FAISS and SLM embeddings with FAISS
- Leveraging Domain-Specific Embeddings for Enhanced Retrieval

**Relevance to Our Team:** This article focuses on hybrid model with SLM which can reduce computational cost and latency of our project compared to LLM based frameworks.

## **MedBioRAG: Semantic Search and Retrieval-Augmented Generation with Large Language Models for Medical and Biological QA**

**Problem Statement:** The paper addresses accuracy and model hallucination issues with medical Q&A of complex and technical literature. The authors built a system called *MedBioRAG* that searches medical documents, finds the most relevant information, uses fine-tuned model and generates answers based on real medical literature.

### **Algorithms Used:**

- *BM25* for lexical matching
- Fine tuned *GPT-4o*

**Datasets:** *NFCorpus*, *TREC-COVID* for text retrieval testing

*MedQA*, *PubMedQA*, *BioASQ* for close-ended QA testing

*LiveQA*, *MedicationQA*, *PubMedQA*, *BioASQ* for long-form QA testing

**Model Training and Testing:** The system was tested on all three types of tasks with respective DBs.

**Results: 1. Text Retrieval** - Semantic search significantly outperformed keyword search.

**2. Close-ended QA** - MedBioRAG achieved higher accuracy than GPT-4o (base), Fine-tuned GPT-4o without retrieval, Previous state-of-the-art models

**3. Long QA** – Responses generated are more coherent and factually correct. Better ROUGE, BLEU, and BERTScore metrics

### **Conclusions:**

- Semantic retrieval improves document ranking.
- Adding retrieval reduces hallucinations.
- Fine-tuned GPT-4o + retrieval works better than GPT-4o alone.

### **Open Questions:**

- Not validated by medical doctors
- Depends heavily on retrieval quality
- Conflicting documents may reduce accuracy
- Computationally expensive
- Not yet tested on real hospital EHR data

**Relevance to Our Team:** MedBioRAG is structurally very aligned with what we are proposing to build.

Reference: Seonok Kim, "Semantic Search and Retrieval-Augmented Generation with Large Language Models for Medical and Biological QA", arXiv, 2025

<https://arxiv.org/abs/2512.10996>

## **Implementation of RAG based Question-Answering Application**

**Problem Statement:** Traditional tools used for reading research lack interactive question-answering capabilities, cannot interpret technical content and require manual reading and analysis of long documents. The proposed system addresses these issues by automatically extracting relevant content, understanding user queries, generating concise, context-aware answers grounded in the document.

### **Algorithms Used:**

- *Streamlit* for frontend
- *Langchain* for Backend Orchestration
- *ChromaDB* for Vector DB
- *Gemini 1.5 Pro* as LLM
- *Google Gen AI embeddings* as embedding model
- *PyPDFLoader* as document loader
- *RecursiveCharacterTextSplitter* for text segmentation
- *Firebase / GCP* for cloud deployment

**Datasets:** User-provided PDF documents like research papers, technical articles, academic documents

**Model Training and Testing:** Instead of a predefined dataset, the system creates its own knowledge base dynamically from uploaded documents. The process is PDF Upload->Text extraction->Text chunking->Embedding creation->Vector DB storage

When a user asks a question:

1. The query is converted into an embedding.
2. The vector database searches for similar document chunks.
3. Relevant chunks are retrieved.
4. The chunks are passed to *Gemini 1.5 Pro* to generate the final answer.

**Results:** There are no standard evaluation metrics used in this paper but all descriptive like Accurate Context-Based Answer, Improved Document Understanding, Efficient Information Retrieval, Reduced Manual Effort, Grounded Responses, Improved User Trusts.

**Conclusions:** RAG architecture is highly effective for document analysis and conversational AI systems.

### **Open Questions:**

- No biomedical dataset
- No quantitative evaluation metrics
- No domain-specific validation
- Focused on general research papers

**Relevance to Our Team:** This work demonstrates a practical implementation of a RAG-based chatbot for document question answering, providing a foundational architecture that can be extended to domain-specific applications such as clinical research literature analysis.

Reference: Suryavanshi, K. et al. (2025) 'Implementation of RAG based question-answering application', ICDSBS.

<https://ieeexplore.ieee.org/document/11031968>

## **Comparative analysis of medical-domain and general-purpose large language models: evaluating specialization in healthcare applications**

**Problem Statement:** Many healthcare AI systems assume specialized medical training is necessary. However, modern general-purpose instruction-tuned models have become very powerful. The study aims to determine:

- Do medical-specific LLMs outperform general LLMs on medical QA tasks?
- Is domain-specific pretraining really required?

### **Algorithms Used:**

**Medical-domain models** - *Meditron-7B, BioMistral-7B, MedAlpaca-13B, PMC-LLaMA-13B*

**General-purpose instruction-tuned models** - *Ministral-8B-Instruct, Gemma-2-9B-it, Vicuna-13B v1.5, Llama-3-8B-Instruct*

**Prompting Strategies** - *Zero-shot prompting, Few-shot prompting*

**Evaluation Metrics** – *BERTScore, SimCSE, Semantic similarity score*

**Datasets:** *PubMedQA, BioASQ, WikiDoc* containing medical research questions and answers derived from biomedical literature.

**Model Training and Testing:** Selected pre-trained LLMs (medical & general), generated 182,944 prompts across datasets, carried out inference testing using zero-shot and few-shot prompts, model outputs were compared with reference answers using semantic similarity metrics.

**Results:** Best Performing Model is *Ministral-8B-Instruct*. Scores - *BERTScore: 0.613, SimCSE: 0.764, Semantic similarity: 0.684*. Best Medical Model is *BioMistral-7B*. Scores- *BERTScore: 0.545, SimCSE: 0.678, Semantic similarity: 0.533*

### **Conclusions:**

- General-purpose LLMs outperform domain-specific medical LLMs.
- Specialized medical training may not be necessary for many medical NLP tasks.
- Model architecture and scale may matter more than domain-specific datasets.
- Zero-shot prompting can be surprisingly effective (eg: *Llama-3-8B-Instruct* achieved *SimCSE = 0.794* in *zero-shot* setting.)

### **Open Questions:**

- Should healthcare AI focus on general LLM improvements instead of specialized models?
- How will results change with larger datasets or newer LLMs?
- Can domain-specific fine-tuning still improve clinical decision support tasks?
- What about safety, hallucination, and clinical reliability?
- How will real clinical workflow integration affect performance?

**Relevance to Our Team:** This work gives insights into model selection, RAG justification, evaluation metrics which is highly useful for our project.

Reference: Roxas, D.Q. and Emekci, H. (2026) ‘Comparative analysis of medical-domain and general-purpose large language models’, *Politeknik Dergisi*.

<https://dergipark.org.tr/en/pub/politeknik/article/1719005>



## **BioMistral-Clinical: A Scalable Approach to Clinical LLMs via Incremental Learning and RAG**

**Problem Statement:** LLMs have shown strong performance in natural language tasks, but general-purpose models struggle in clinical applications due to lack of domain-specific medical knowledge, inability to adapt continuously to new clinical data, high computational costs of training large models, lack of up-to-date contextual knowledge during inference, difficulty in handling long unstructured clinical notes.

Therefore, the paper aims to develop a lightweight clinical LLM system that can learn continuously from unstructured clinical notes, improve clinical reasoning and question answering, use retrieval of external medical knowledge and maintain computational efficiency. The proposed solution is the BioMistral-Clinical system, which combines Incremental learning, Prompt engineering, Retrieval-Augmented Generation (RAG)

### **Algorithms Used:**

**Prompt Engineering** – *Zero-shot, Few-shot, Chain-Of-Thought(CoT) prompting*  
*SPeCiaL(Self-Supervised Training for Continual Learning) for continual learning*

*LoRA (Low Rank Adaptation) for clinical question answering tasks*

**RAG** - *Maximum Inner Product Search (MIPS) to enhance model responses*

**Datasets:** *Clinical Training Dataset, Augmented Clinical Notes Dataset* containing symptoms, diagnosis, treatment, patient outcomes. *MedQA, MedMCQA* for evaluation

**Model Training and Testing:** Data Structuring -> Incremental Pretraining -> SFT -> RAG

**Results:** Compared with the base model, BioMistral-Clinical produces more detailed clinical answers, uses specific medications and treatments, shows better contextual reasoning, avoids answering when insufficient medical data is available.

**Conclusions:** The proposed system demonstrates that efficient clinical LLMs can be built without extremely large models.

### **Open Questions:**

- Limited Clinical Evaluation
- Dataset Constraints
- Safety and Reliability
- Scalability of RAG systems

**Relevance to Our Team:** This paper is **highly relevant** because it demonstrates almost the same architecture as our capstone project.

Reference: Chen, Z. et al. (2025) ‘**BioMistral-Clinical: A scalable approach to clinical LLMs via incremental learning and RAG**’, *Findings of IJCNLP-AAACL 2025*.

<https://aclanthology.org/2025.findings-ijcnlp.71/>

## **Building Customized Chatbots for Document Summarization and Question Answering using Large Language Models using a Framework with OpenAI, Lang chain, and Streamlit**

**Problem Statement:** Traditional search systems require manual reading of large documents, do not provide contextual answers and are inefficient for knowledge extraction.

The paper proposes a customized chatbot framework that can automatically summarize large documents, answer user questions based on document content, provide an interactive user interface, allow developers to easily build document-based chatbots. It integrates LLMs, document processing, and web UI technologies to create an end-to-end chatbot system.

### **Algorithms Used:**

**LLMs** – *OpenAI GPT* models, *RAG QA*, *Recursive Text Splitting* for document chunking, *semantic embeddings*, *prompt engineering*, *LangChain* for orchestrating the pipeline, *Streamlit* to build web application interface

**Datasets:** User provided documents

**Model Training and Testing:** The system uses pretrained LLM APIs.

**Results:** Accurate Document Summarization, Contextual Question Answering, User-Friendly Interface, Practical Implementation

**Conclusions:** The framework demonstrates an effective approach for building document-based conversational systems.

### **Open Questions:**

- Dependence on External APIs
- Scalability Challenges
- Data Privacy
- Limited Evaluation Metrics

**Relevance to Our Team:** This paper is directly relevant, especially for the frontend and system architecture.

Reference: Pokhrel, S. et al. (2024) '**Building customized chatbots for document summarization and question answering using large language models**', *Journal of Information Technology and Digital World*.

<https://irojournals.com/itdw/article/view/6/1/6>